

Ka-band Stripline One-to-Sixteen Corporate Feed Network Using Wilkinson Power Dividers For a 4x4 Phased Array Built-In Self-Test System

Russell Anderson
Electrical and Computer Engineering
University of Massachusetts Amherst
Merrimack, USA russellander@umass.edu

This paper presents a viable option for a one-to-sixteen corporate feed network for a 4 x 4 phased array system. Using Wilkinson power dividers, the corporate feed network is designed and optimized to have the lowest variability between each of the sixteen ports. The feed network is a stripline design utilizing industry leading dielectrics and thin-film resistor technology for a wideband range of 27-40 GHz.

I. INTRODUCTION

Active electronically scanned arrays (AESA) require a lattice of radiating elements to electronically steer a beam at a given angle. Element spacing is crucial to avoid grating lobes which can ruin phased array's performance. This becomes increasingly challenging as millimeter wave technology is becoming more prevalent. This forces designers to consider using multi-board level technology to layout all the front and back-end electronics within the confined space needed for millimeter frequencies. Recent trends in active phased array applications require a wide-bandwidth, low loss corporate feed network in order to properly feed the radiating elements. A corporate feed network is one with equal length branches to all nodes in the system. This corporate feed network was designed for stripline in-between PCB board layers for minimal impact on overall array size as well as adhering to element spacing obligations. Low dielectric constant (ϵ_r) and loss tangent ($\tan \delta$) are desired with high-frequency circuits.

When considering dielectric substrates for Ka-band designs, Tachyon® 100G substrates from Isola® is a leading industry standard. In order to create a stripline stack up, two layers of 0.051mm thick Tachyon® 100G substrate are used. These layers have a dielectric constant of 2.97 and a loss tangent of 0.0014. The use of stripline instead of microstrip is due to the improved shielding of electromagnetic waves through the second top layer of dielectric material and ground cover plate. This will help reduce insertion loss as well as improve isolation between ports of the feed network.

In order to properly produce an equal split of transmit and receive signals flowing through the phased array system, Wilkinson power dividers are used. Properties of the Wilkinson are desirable when considering a corporate feed network due to the symmetry of the divider and the isolation characteristics. Ideally the signal is applied to the common port of the Wilkinson and is split evenly into two signals with half the power. Wilkinson power dividers use quarter-wave lines to provide an

impedance transformation from one common-port to two radiating ports. The two radiating ports are connected by a resistor with a value of two times the system impedance (Z_o). Ticer™ Technologies provides thin-film embedded resistor technology embedded into the copper foil. This allows the stripline design to be embedded into board layers with minimal impact on space. A series of sixteen Wilkinson power dividers is used to synthesize a corporate feed network with a 4 x 4 pattern. Each of the sixteen radiating ports are equally spaced to the highest frequency of interest using

$$d = \frac{\lambda}{2} \quad (1)$$

where d is the distance between elements and λ is the wavelength of the highest frequency of interest. Feed lines are routed to form an H-Tree pattern as seen in Fig. 1. The corporate feed should have a magnitude variation between each radiating port of no more than 0.5dB and a phase variation between each port of no more than $\pm 5^\circ$. This ensures that the variation from port to port does not affect the overall system performance. This also allows for proper calibration of the system. This feed network can also be used as a built-in self-test (BIST) system to calibrate the phased array system while in the field. BIST systems will sample the signals being either transmitted or received to calibrate the phase shifters, variable attenuators, and other

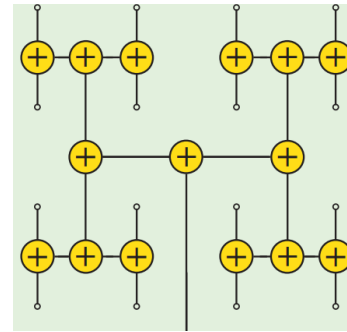


Fig.1. 1-to-16 4 x 4 H-Tree Pattern

Ref: Vouvakis, M., Lecture Notes 4. Behind Ground Sub-Systems, slide 197

beamforming components in the system. Proper calibration ensures that the system pushes the correct beamforming data to

the required components which will allow for accurate detection and tracking.

II. WILKINSON POWER DIVIDER

A. Component Design

A Wilkinson power divider is composed of a series trace line with impedance of Z_0 (50Ω) and an electrical length equal to 0.25λ . Two branches are formed by a T-junction of the series line. These branches are designed to have impedance of $\sqrt{2}Z_0$ with the same electrical length of 0.25λ . The two branches are then transitioned back to 50Ω lines to present a good match to the next Wilkinson or connector. Line dimensions are determined by a CAD line calculator using the Tachyon® 100G dielectric constant and loss tangent values. Board thickness is determined to be 0.128mm which will dictate the trace widths with respect to specified impedance values. The electrical length dictates the trace length as the TEM wave travels through the stripline. The line dimensions were calculated for a center frequency of 33.5GHz to accommodate 27 – 40 GHz.

The two branches of the Wilkinson power divider are connected by a resistor with impedance equal to twice Z_0 (100Ω). Dimensions of this resistor and material selection is dictated by the required power handling at that point in the chain. Typically beamforming signals are not considered high power which allows for smaller form-factor components. Ticer™ Technologies TCR® technology offers a thin-film copper resistive foil which is applied as a sheet between the two branch ports. This copper process is controlled for uniform composition and deposition thickness which ensures consistent resistance values. The resistance will vary to a slight degree of less than $\pm 5\%$.

B. Specifications and CAD EM Simulation

The Wilkinson power divider should have a good input return loss (S_{11}) and branch isolation (S_{23}) of at least -10dB. The insertion loss (S_{21} , S_{31}) of the branch ports should be an equal split of power at -3dB down with parasitic losses accounted for. As stated, the variation in insertion loss should not exceed 0.5dB. A CAD simulation is performed using Keysight ADS and Momentum EM. Fig.3 shows the response of an ideal Wilkinson using a lumped-element simulation of ideal stripline components and dielectric board parameters. In Fig.2 a Wilkinson power divider was then designed using realistic stripline geometries which suited the design specifications and allowed for compact element spacing. The design was simulated using Keysight's Momentum full electromagnetic wave simulation which creates a series of mesh to solve boundary condition equations of electric and magnetic fields and waves. Fig.3 and Fig.4 shows the Momentum simulation results along with the ideal Wilkinson response. There is a clear difference between the ideal Wilkinson response and the more realistic full-wave response, however, the design specifications for return loss, isolation, and insertion loss are still met across 27 – 40 GHz.

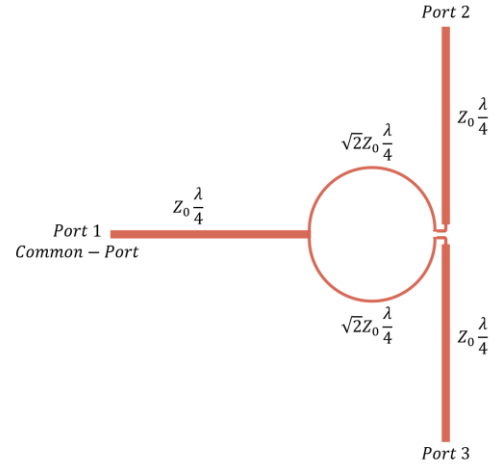


Fig.2. 1-to-2 Stripline Wilkinson power divider with impedance and electrical length values.

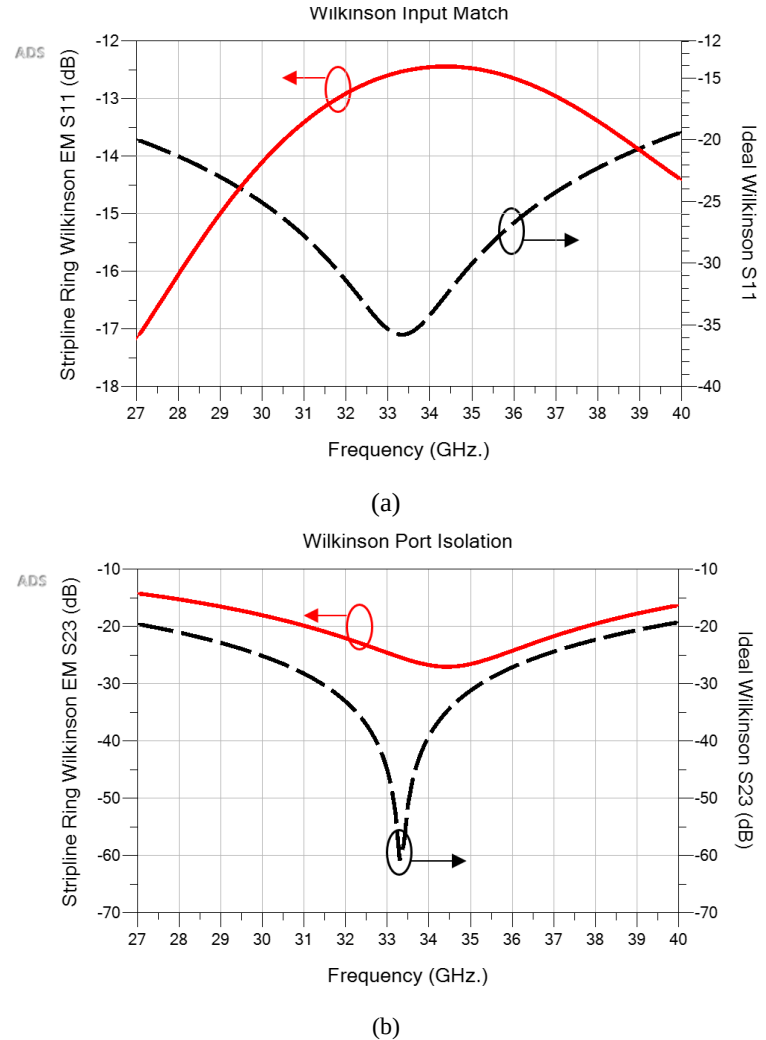


Fig.3. Wilkinson ideal and Momentum EM simulation results. (a) S_{11} input match. (b) S_{23} Branch port isolation

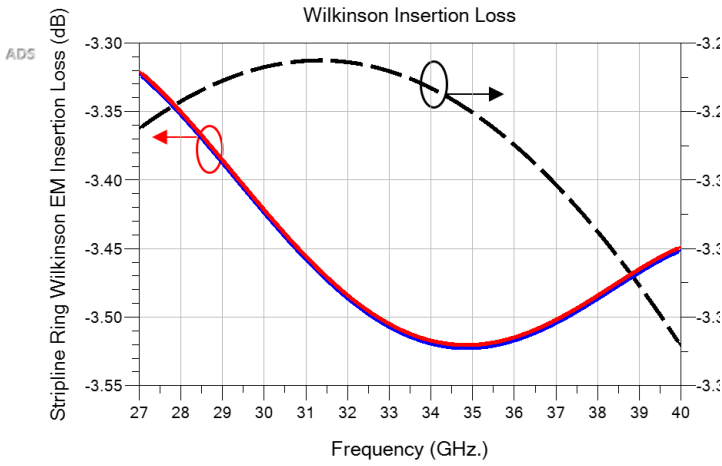


Fig.4. Wilkinson ideal and Momentum EM S_{21} insertion loss.

III. CORPORATE FEED NETWORK

A. Design Specifications and Requirements

As stated in the introduction, the purpose of the feed network is to feed the antenna or radiating elements of the phased array system. The radiating elements must be spaced according to the highest frequency of operation. Symmetry in these systems is crucial to help eliminate amplitude and phase errors between each channel. This still holds true for a BIST calibration system as to provide accurate calibration data to components. The required frequency range of 27 – 40 GHz. means an element spacing of 3.75mm is necessary. Wilkinson power dividers are arranged such that each of the 16 ports are evenly spaced by this amount both in the x and y dimension. A 4 x 4 lattice is formed by routing the Wilkinson power dividers from the common port.

Since there are four Wilkinson power splits to each radiating port from the common port, there ideally should be a 12dB of insertion loss. The corporate feed network should have input/output return loss and port to port isolation of at least -10dB. Fig.6 shows the full corporate feed network using stripline Wilkinson power dividers and transmission lines. This feed network was designed and simulated in Keysight Momentum to build the full wave mesh and then co-simulated with lumped element resistors due to the difficult nature of simulating non-traditional technology of the Ticer™ TCR® thin-film resistors.

B. CAD EM Simulation and Results

First, an ideal corporate feed system is simulated using the ideal lumped element simulation for the Wilkinson power divider. This is used to compare the full-wave EM simulation from Momentum. Results are compared against the design specifications. To better provide an accurate model of the system, a Monte-Carlo simulation is performed to simulate the variability of the resistor value which is given by the vendor as $\pm 5\%$. Fig.5 depicts this variability in the resistance. The Monte-Carlo simulation was run for one thousand runs while randomly varying the resistance with a Gaussian distribution. All small-signal responses for the corporate feed network show the variation of the resistor value for each Monte-Carlo

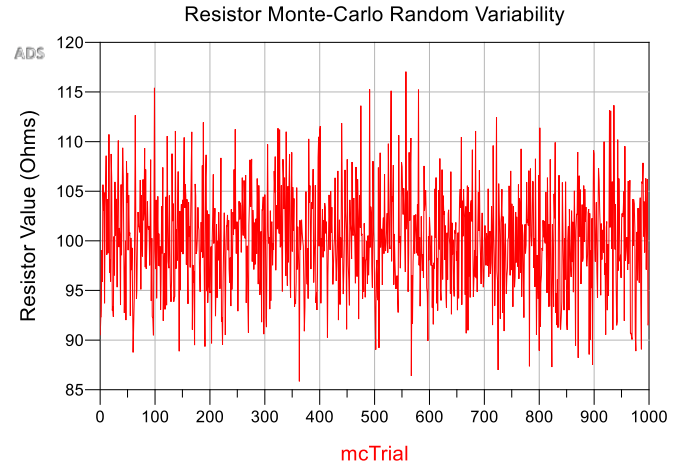


Fig.5. Monte-Carlo simulation of Ticer™ TCR® thin-film resistor variation plotted vs. trial.

sweep. This can be seen in Fig.7 where the swept curves appear in red with a blue line to indicate where the resistor value is exactly 100Ω. Analyzing Fig.7 the design specifications for input and output return loss, port to port isolation, as well as the variation in signal magnitude and phase. The insertion loss can be seen ranging from -13dB to -13.4dB. The additional loss of 1dB is due to the cascaded parasitic losses of the transmission lines. Fig.7 also shows all 16-port phase across the entire frequency band. It can be noted that each of the one thousand Monte-Carlo simulations is also plotted, however, since there is little variation in the phase as the resistor changes the curves appear on top of each other. Dashed lines represent the ideal corporate feed network results. It can be observed that the phase variation is with the specifications of $\pm 5^\circ$.

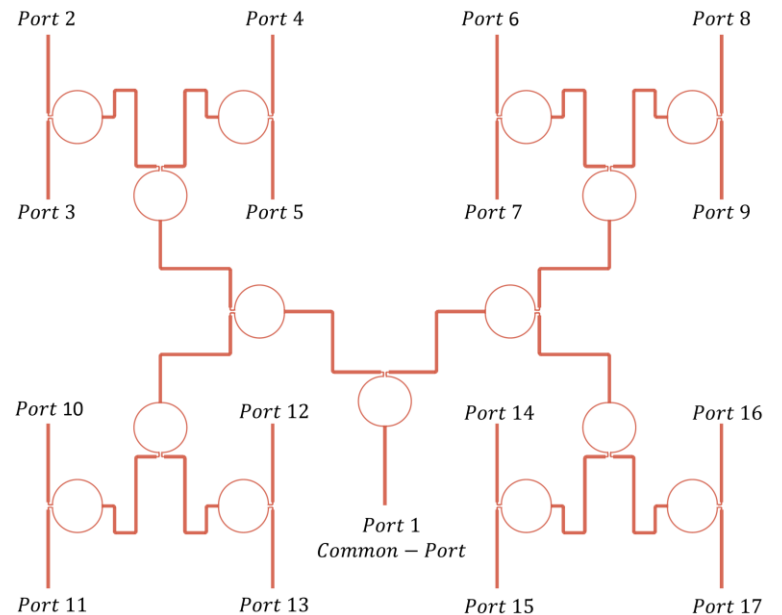
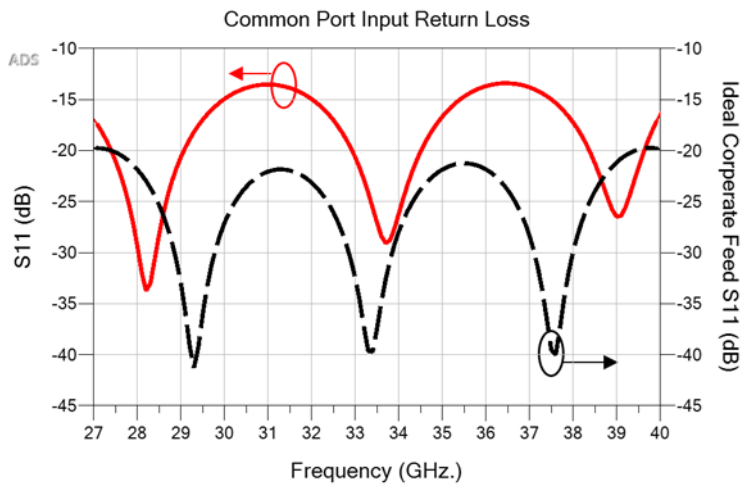
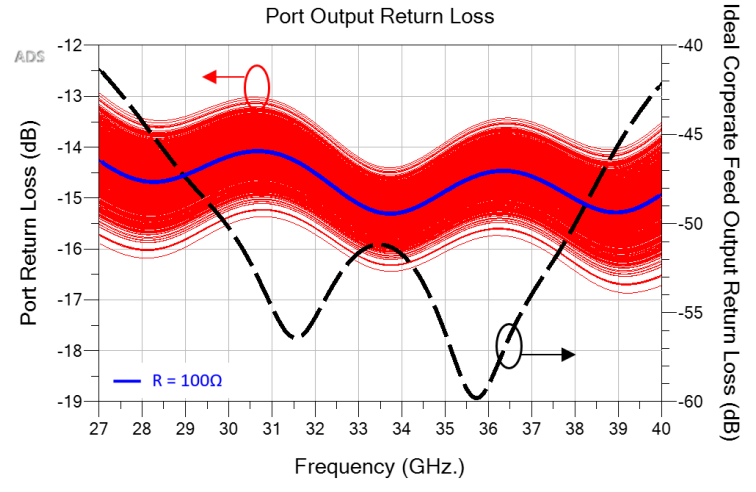


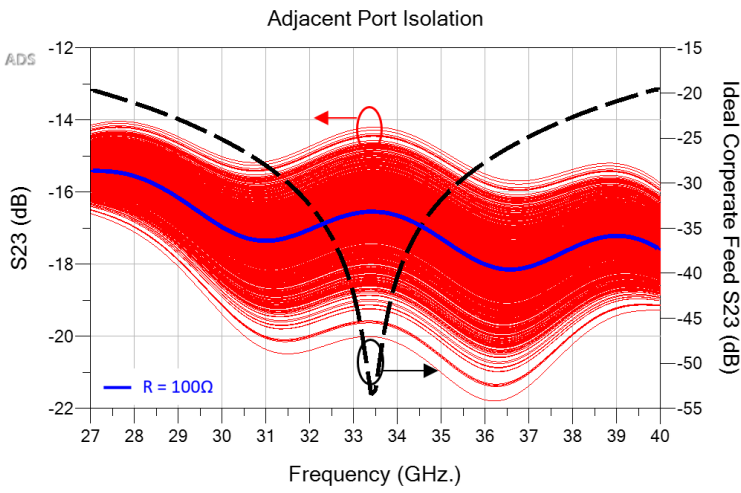
Fig.6. 4 x 4 1-to-16 Corporate feed network.



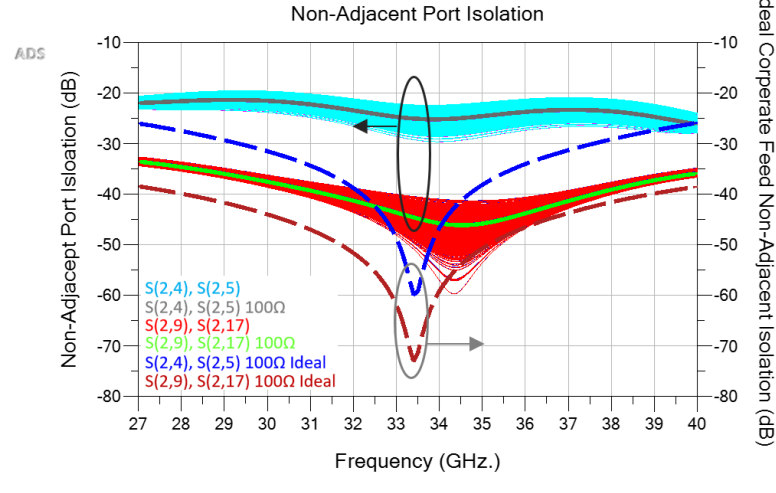
(a)



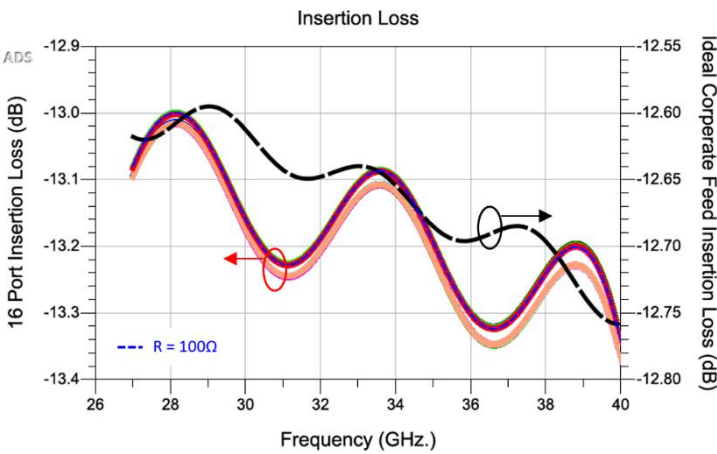
(b)



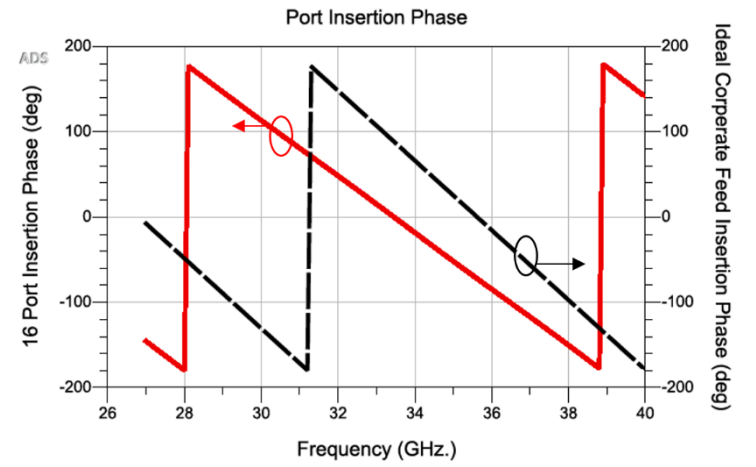
(c)



(d)



(e)



(f)

Fig.7 Ideal and Momentum EM simulation results of corporate feed network. (a) Input return loss S_{11} . (b) Output return loss S_{22} . (c) Adjacent port isolation S_{23} . (d) Non-adjacent port isolation. (e) 16 port insertion loss. (f) 16 port insertion phase.

As shown in Fig.7, the input match, output match, and isolation are all below the specified -10dB across the entire 27 – 40 GHz. band. While the variation in the resistor does not seem to affect the input return loss, insertion loss and phase, the output return loss and port isolation change drastically. For higher levels of resistance, the more isolation each port has along with better output return loss. This is due to the increase in impedance seen at each port. Accounting for the worst variation in resistance, the port-to-port isolation is better than -14dB and the output return loss is better than -13dB.

IV. CONCLUSIONS

This paper offers a solution to the design and simulation of a one-to-sixteen corporate feed network for a 4 x 4 phased array system operating over the frequency range of 27-40 GHz. The feed network implements this using Wilkinson power dividers in order to obtain an even power split to 16 ports with port-to-port insertion loss variation of less than 0.1dB and less than 0.5dB variation over the entire frequency range. Similarly, the phase variation is less than 0.1dB from port to port for all 16 ports. Monte-Carlo analysis is crucial for determining affect of resistor variability within Wilkinson power dividers within the feed network. This builds a best- and worst-case scenario for the network performance. To better model the system, surface roughness should be accounted for. This will allow for a more accurate insertion loss result at higher frequencies were crucial. Both the Tachyon® 100G and Ticer™ TCR® thin-film resistor surface roughness should be accounted for. To optimize the simulation, the feed network should be modeled with radiating elements in a 3D CAD software package such as Ansys HFSS. This way you could quantify the beamforming performance of the feed network and allow for any design modifications. Once designed the feed network could be fabricated, tested, and further optimized for desired performance. Once the final design is achieved its use as a built-in self-test system can be optimized with the use of calibration weights through the beamforming algorithm.

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